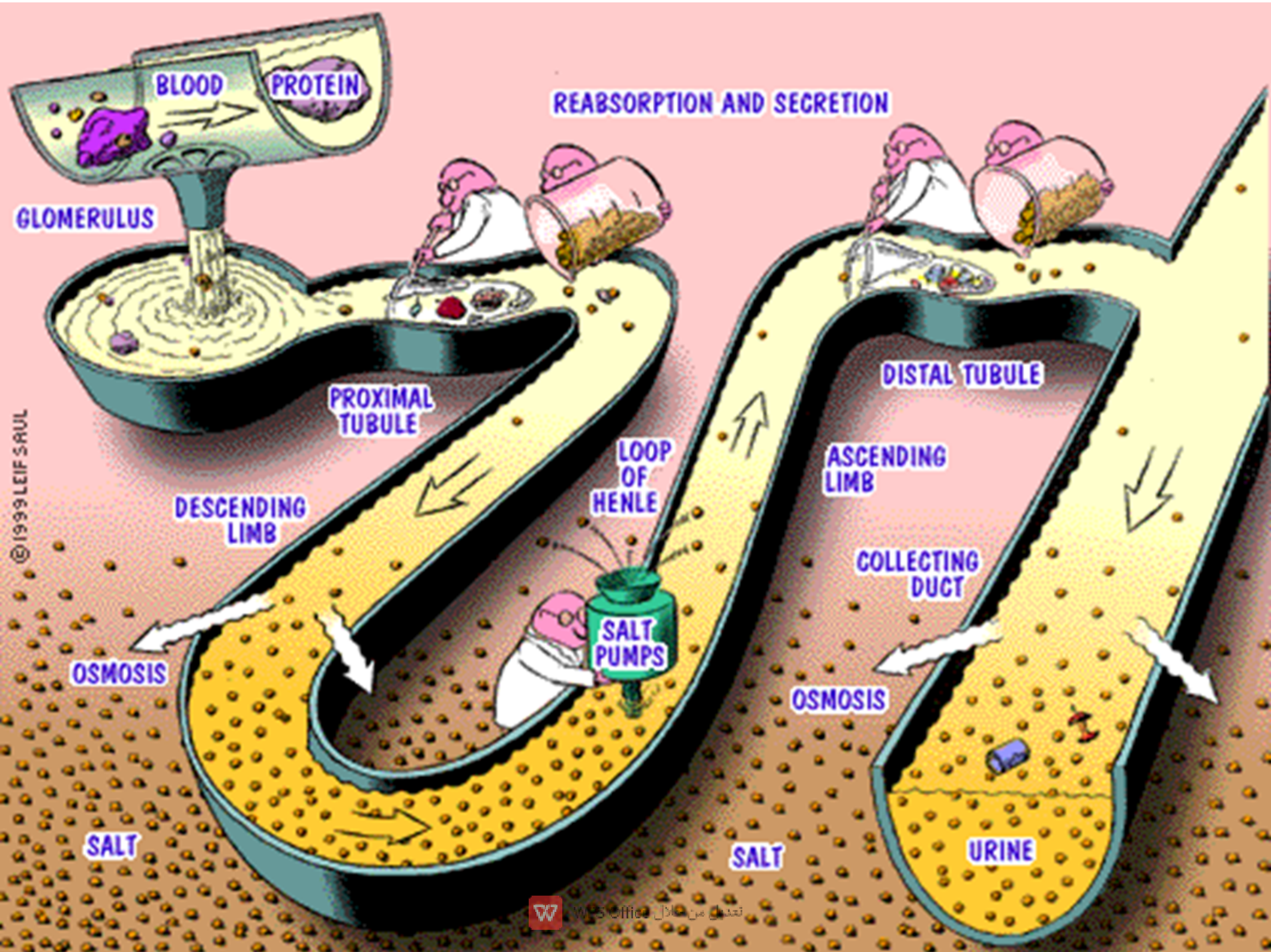




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Concentration of Urine

- Osmolarity of glomerular filtrate is same as that of plasma and it is 300 mOsm/L.
- Normally urine is concentrated and its osmolarity is four times more than that of plasma, i.e. 1,200 mOsm/L.
- Osmolarity of urine depends upon two factors:
 - 1. Water content in the body
 - 2. Antidiuretic hormone (ADH).



FORMATION OF CONCENTRATED URINE

- It involves two processes:
- 1. Development and maintenance of medullary gradient by countercurrent system
- 2. Secretion of ADH.



COUNTERCURRENT MULTIPLICATION

countercurrent multiplication refers to the ability of the kidney to increase the osmolality of the interstitium from the normal value of 300 mosm/kg up to 1200–1400 mosm/kg

this is necessary to allow the kidney to be able to concentrate the **urine**



Divisions of Countercurrent System

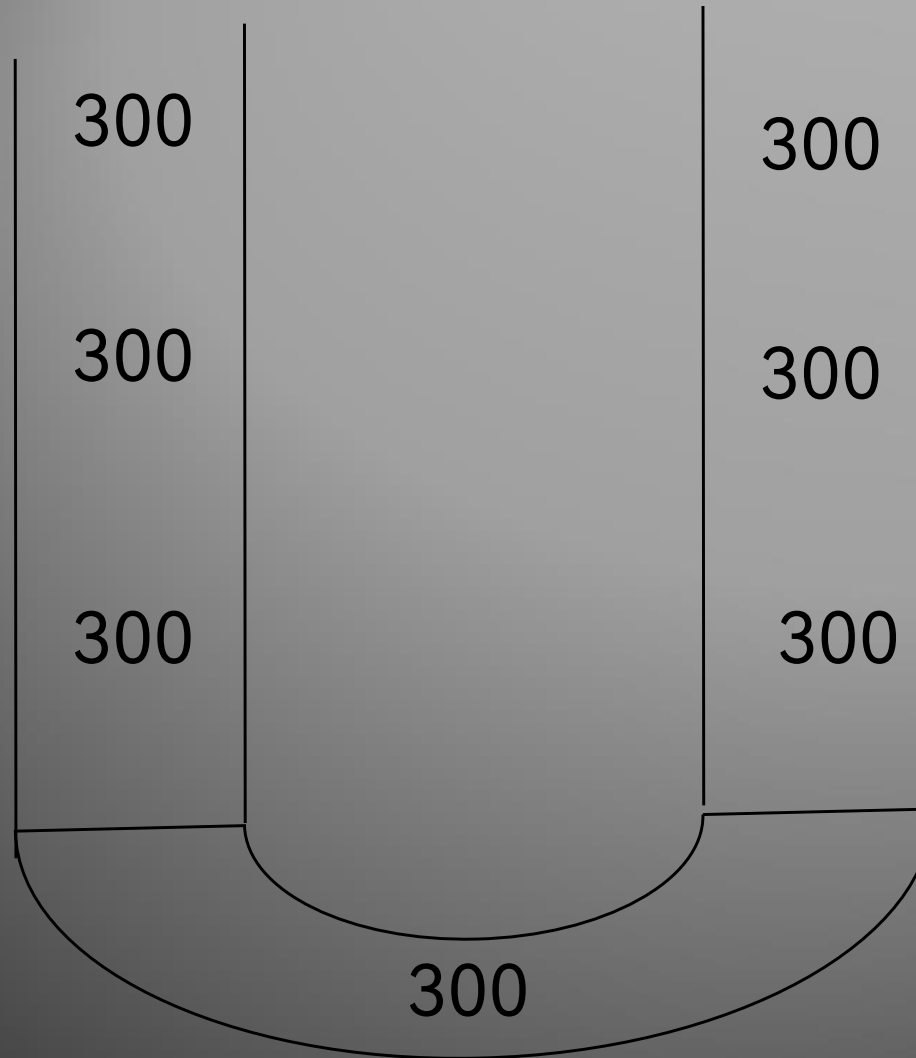
Countercurrent system has two divisions:

1. Countercurrent multiplier formed by loop of Henle
2. Countercurrent exchanger formed by vasa recta.



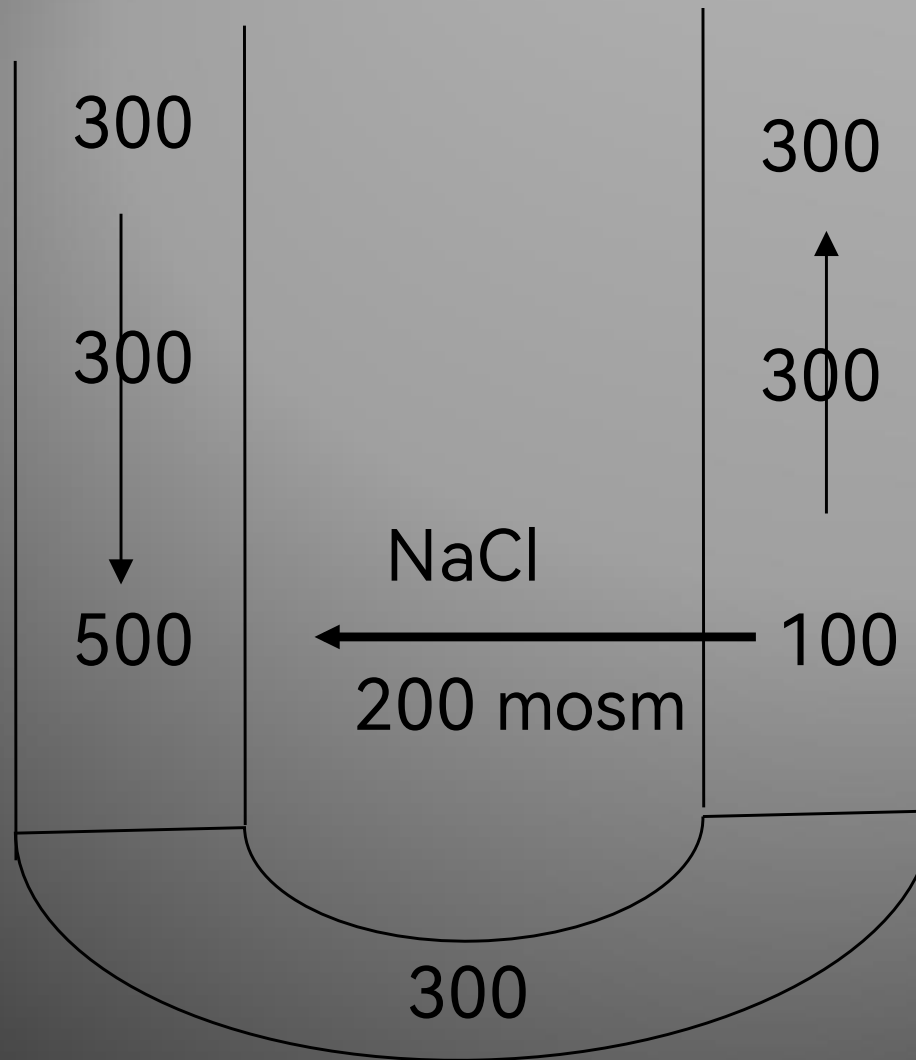
COUNTERCURRENT MULTIPLICATION

TIME 0



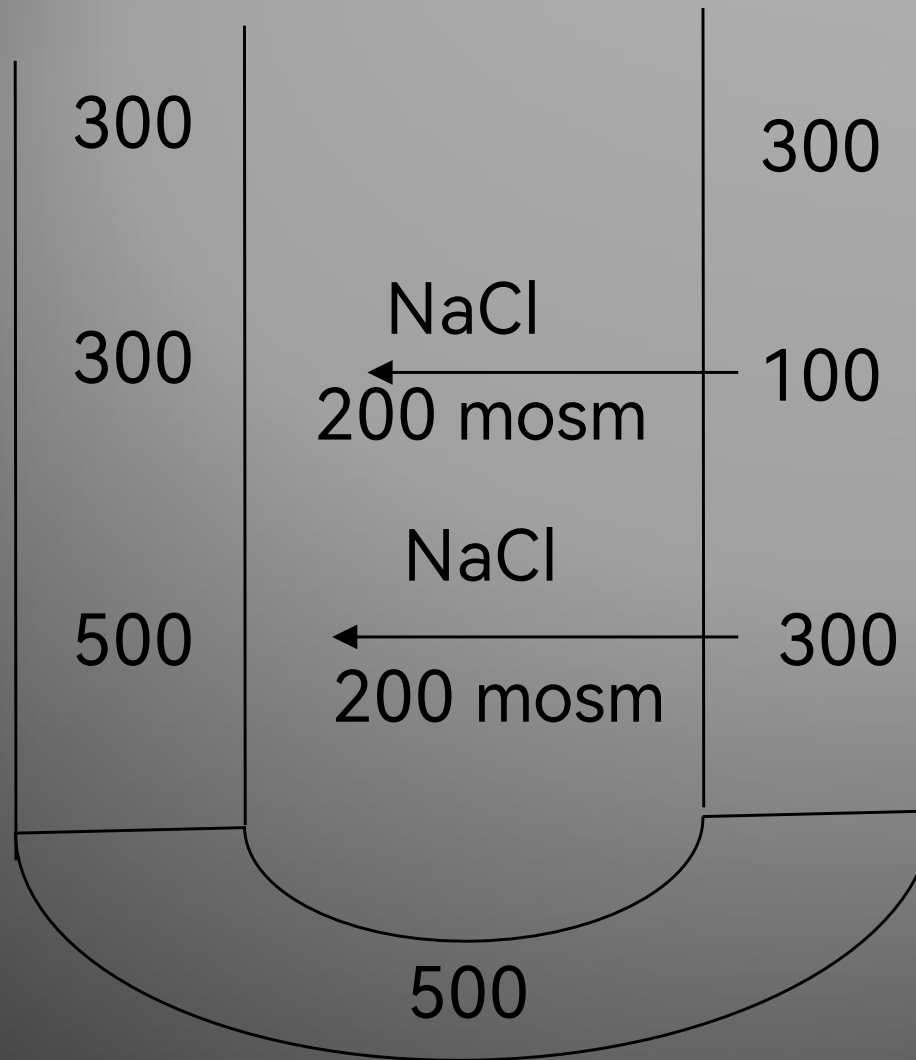
COUNTERCURRENT MULTIPLICATION

TIME 1



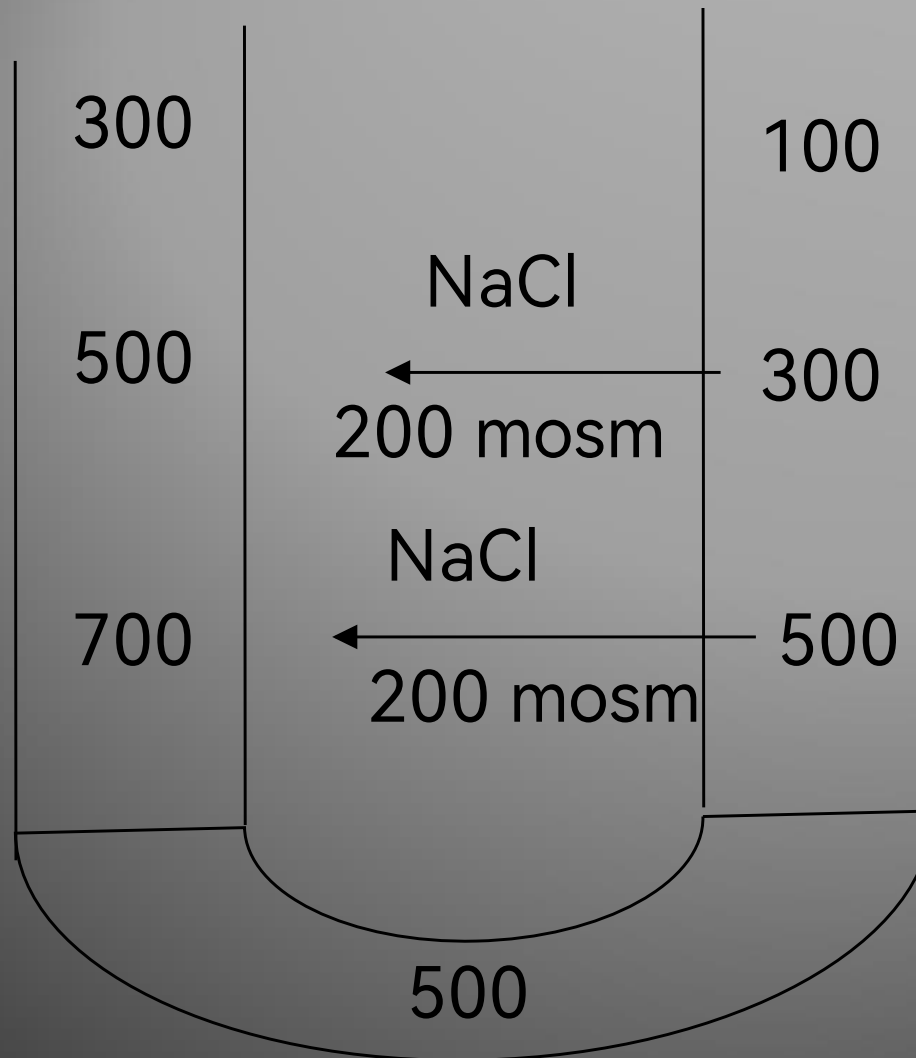
COUNTERCURRENT MULTIPLICATION

TIME 2



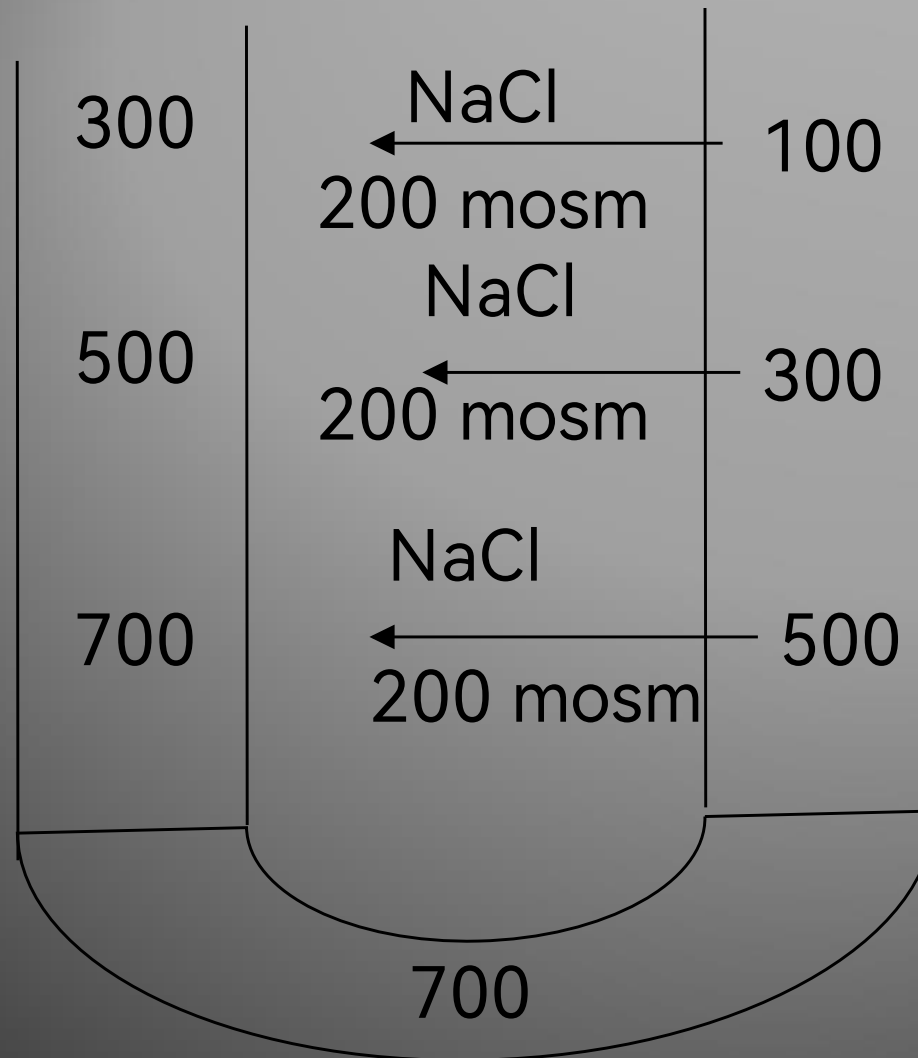
COUNTERCURRENT MULTIPLICATION

TIME 3



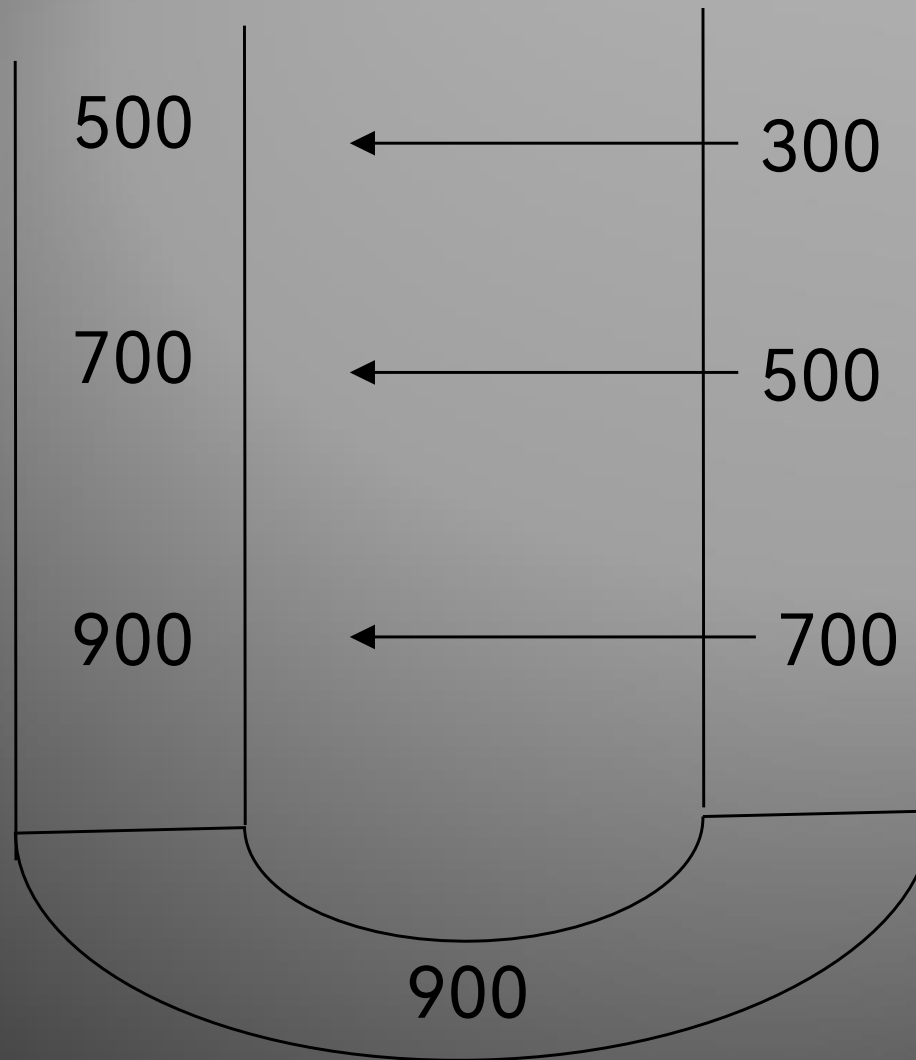
COUNTERCURRENT MULTIPLICATION

TIME 4



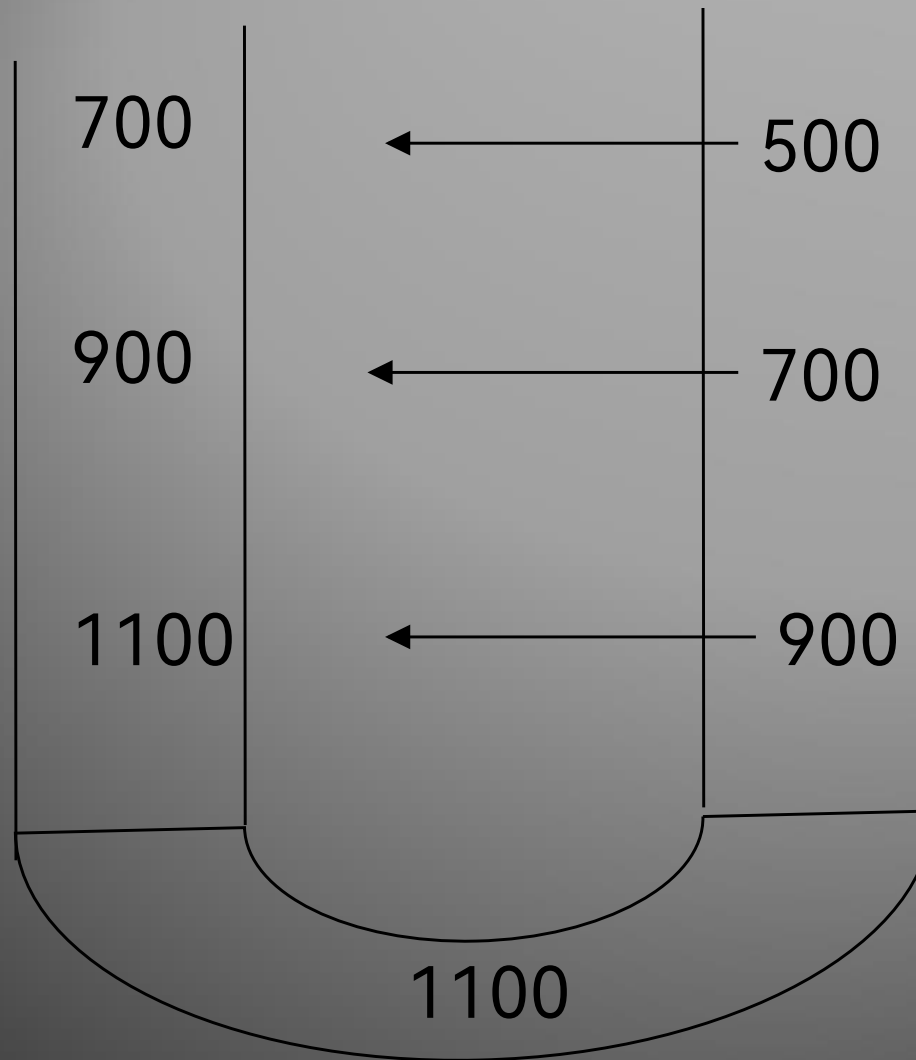
COUNTERCURRENT MULTIPLICATION

TIME 5



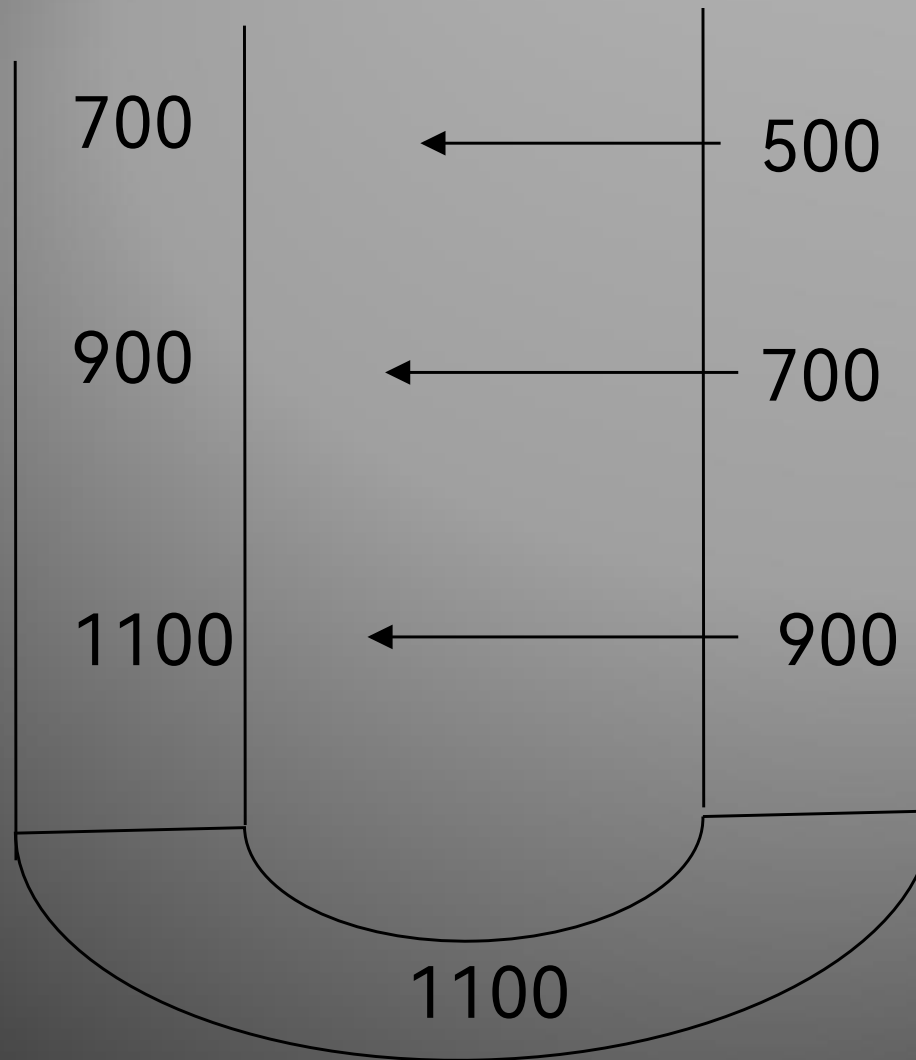
COUNTERCURRENT MULTIPLICATION

TIME 6

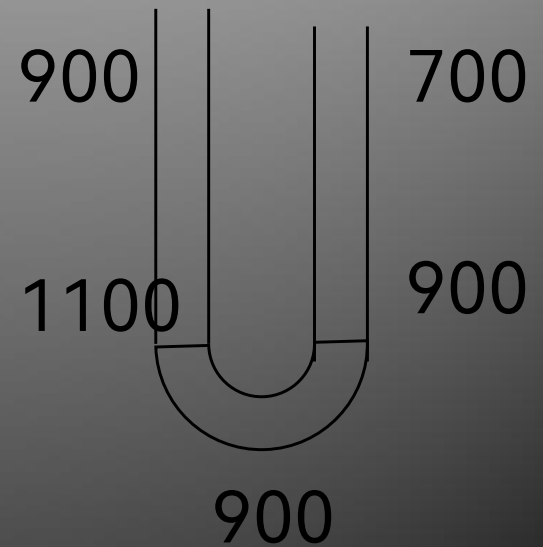
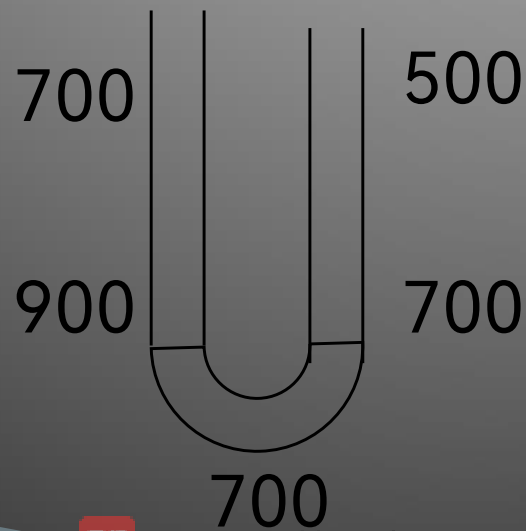
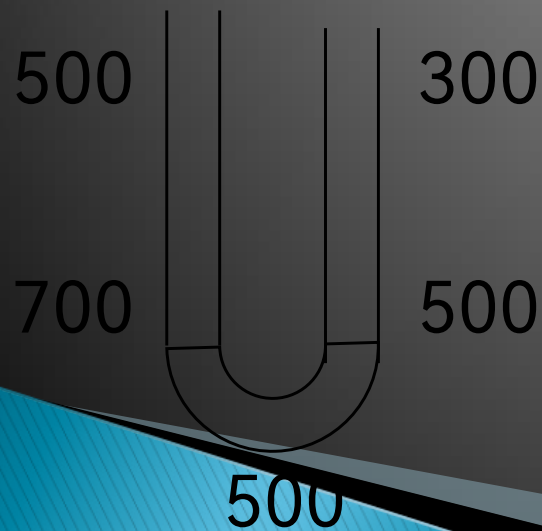
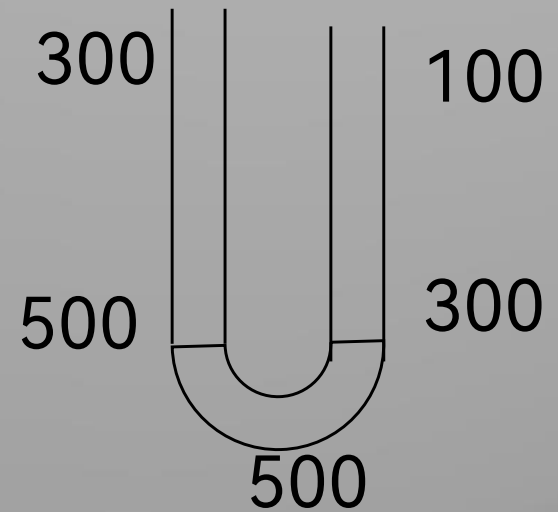
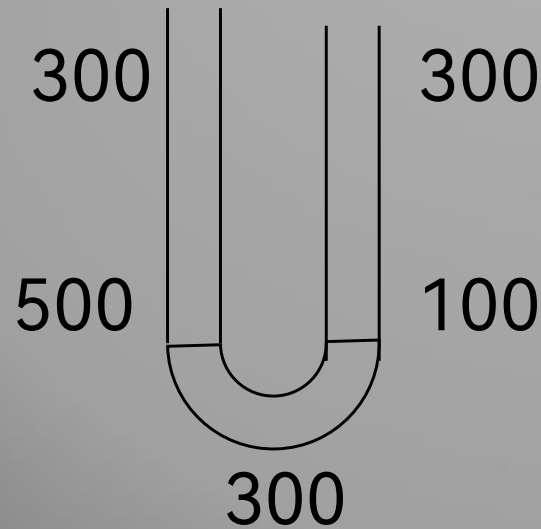
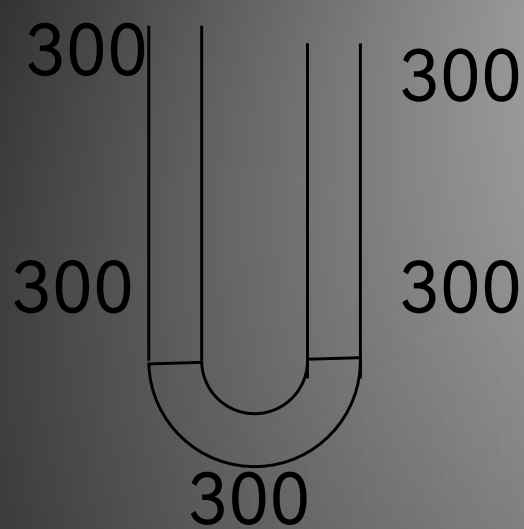


COUNTERCURRENT MULTIPLICATION

TIME 7

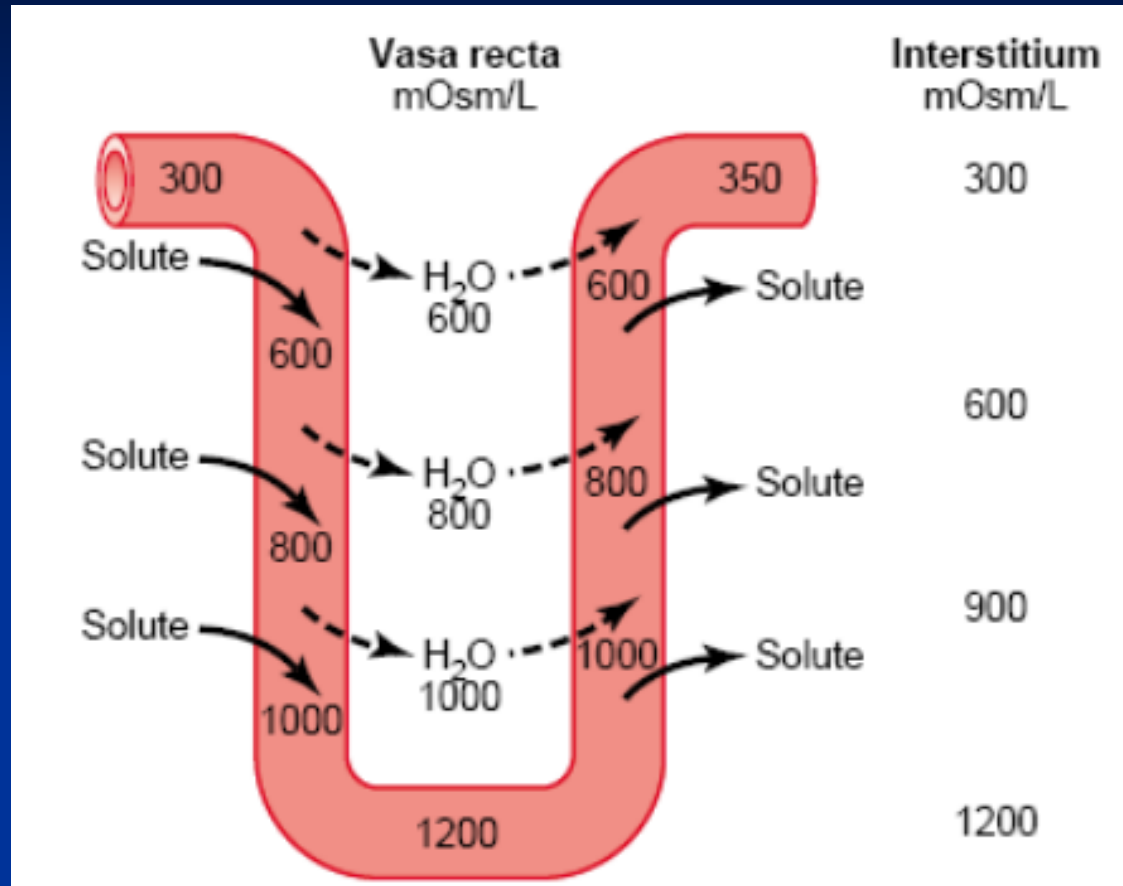


COUNTERCURRENT MULTIPLICATION



The countercurrent exchange

The vasa recta



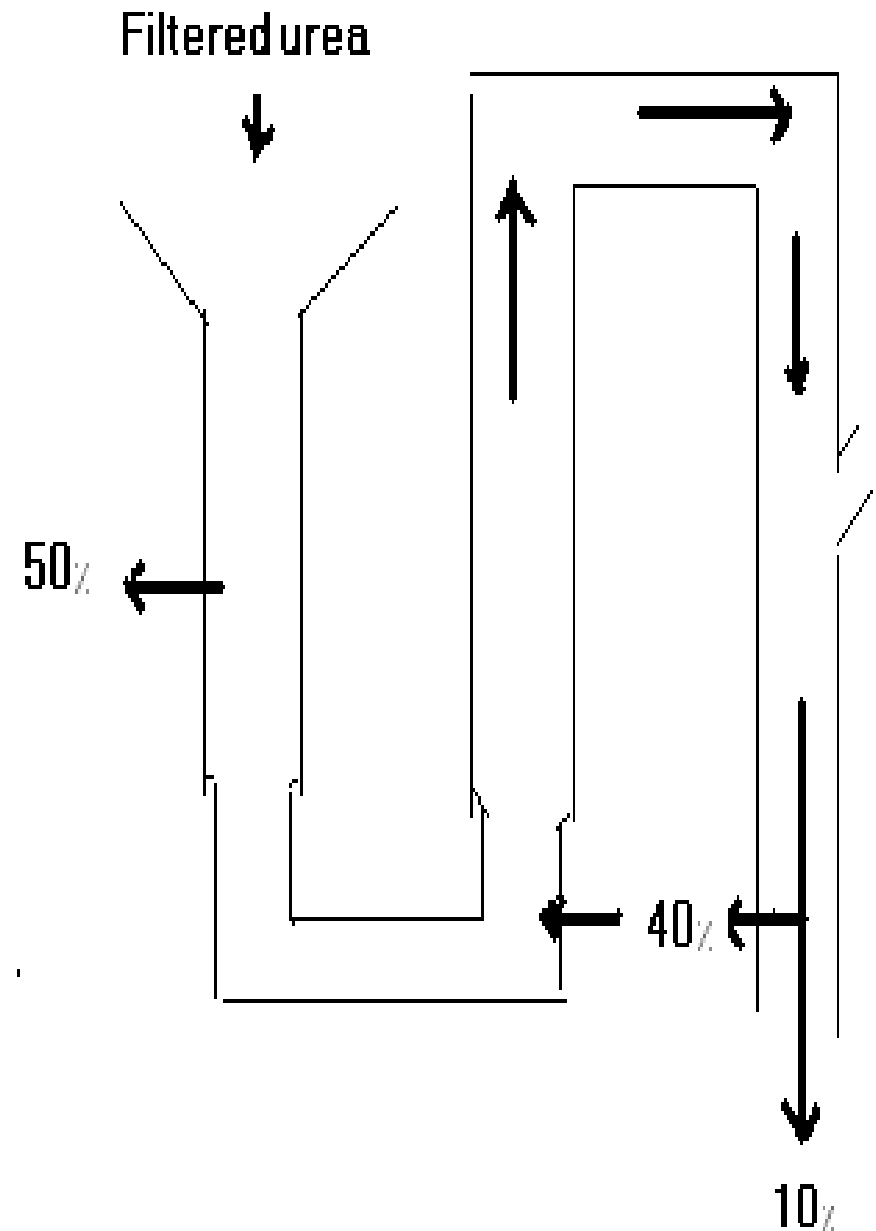
- In the descending limb, water diffuses out of the blood and solutes diffuse from the renal interstitial fluid into the blood.
- In the Ascending limb, the opposite occurs.
- This process **preserves** the hypertonicity of the medullary interstitial fluids.

Role of urea

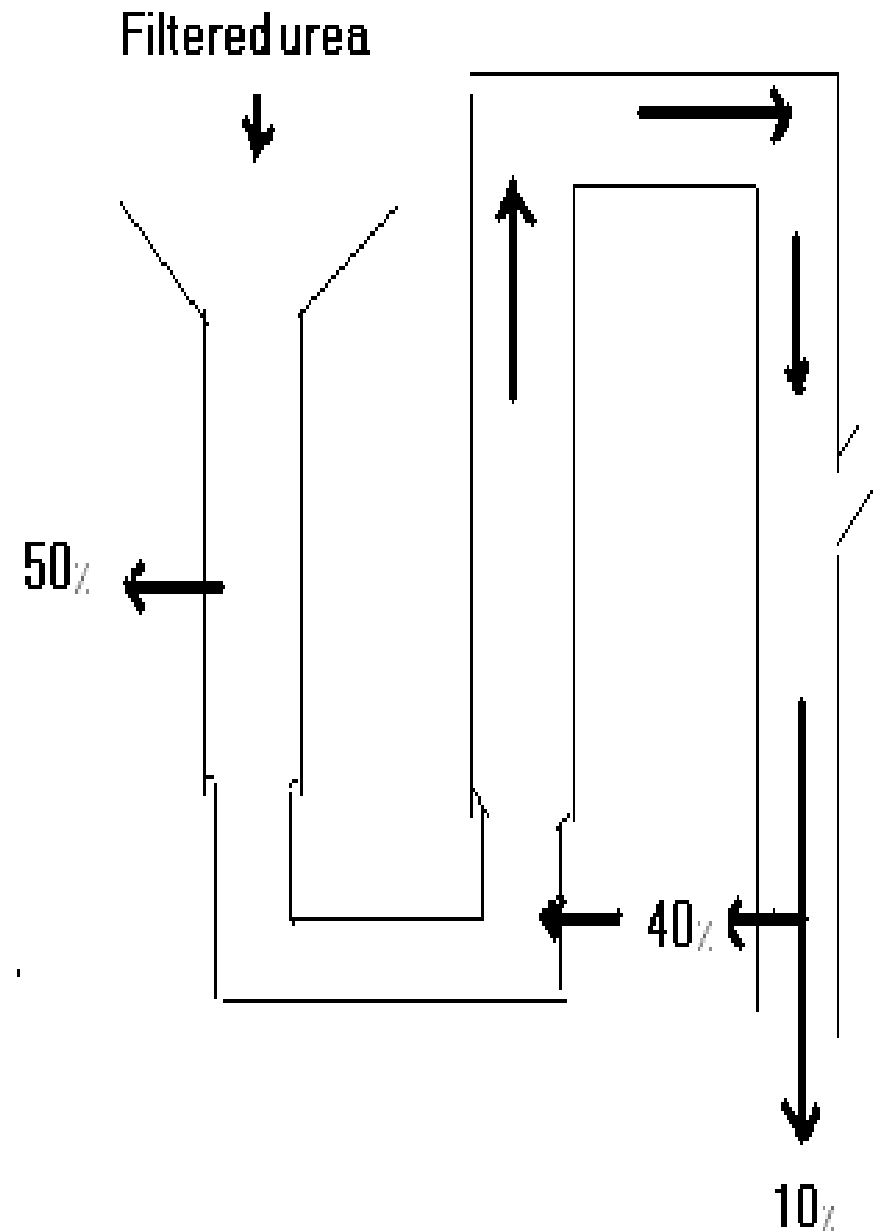
- A high protein diet increases the ability of the kidney to concentrate urine
- Urea recycles between the tubules and the interstitium and so increases hypertonicity of the medulla



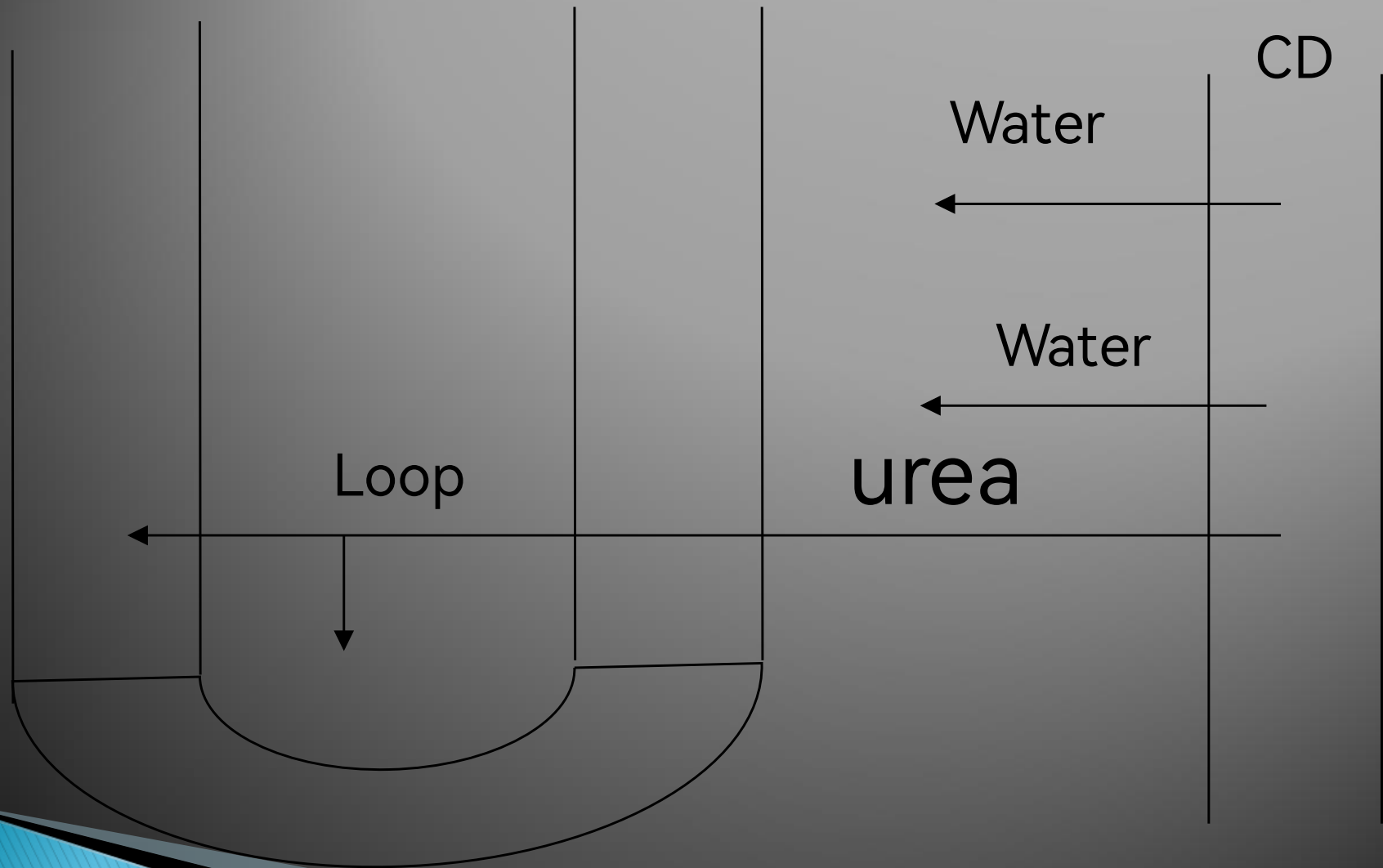
- The amount filtered:
 - depends on the amount of protein diet
 - 50% is reabsorbed in the PCT
 - 10% is excreted in urine



- 40% is reabsorbed in the collecting ducts to the interstitium (facilitated by ADH)
- This enters the thin ascending limb of the loop of Henle & recycles between the interstitium & the tubules



THE ROLE OF UREA IN COUNTERCURRENT MULTIPLICATION



Any QUESTION

THANK YOU

